

Background

Little Red Dots (LRDs) are compact, red sources seen by JWST, often interpreted as active galactic nuclei. Their spectra show broad Balmer lines and blackbody-like continua, but their nature is debated.

LRDs show features unlike any known AGN, and inferred black hole masses seem too high for their host galaxies.

What this paper is selling

This paper draws parallels between LRDs and stellar eruptions like eta Carinae and Type IIIn supernovae. It proposes that LRDs are super-Eddington, intermediate-mass engines enshrouded in dense gas, not overmassive black holes.

Key results to remember

- LRDs cluster at ~5000 K, like Type IIIn SNe and eta Carinae.
- Wind-shocked pseudo-photospheres explain LRD spectra and line profiles.
- Engine masses are likely 10³⁻⁶ solar masses, not supermassive.

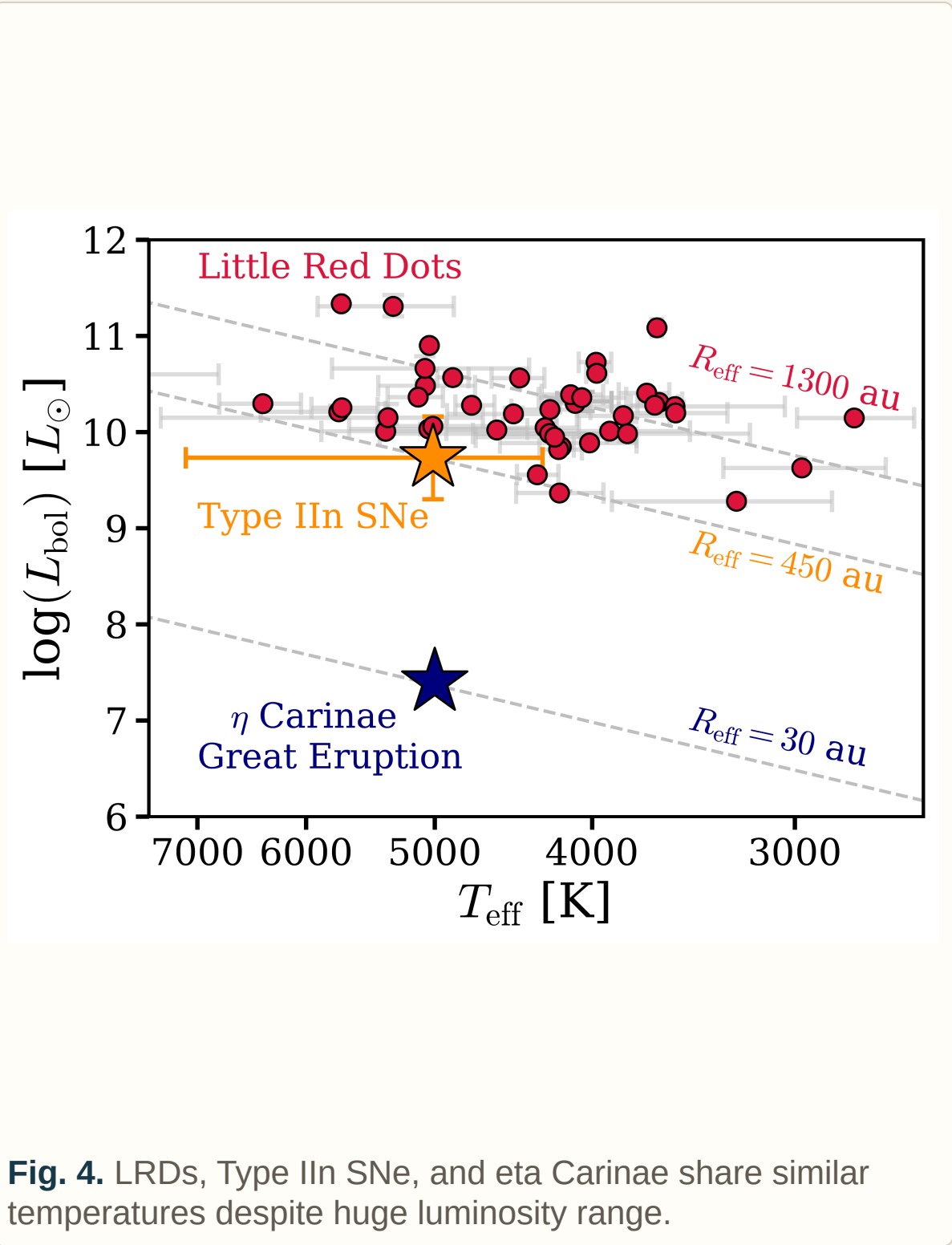


Fig. 4. LRDs, Type IIIn SNe, and eta Carinae share similar temperatures despite huge luminosity range.

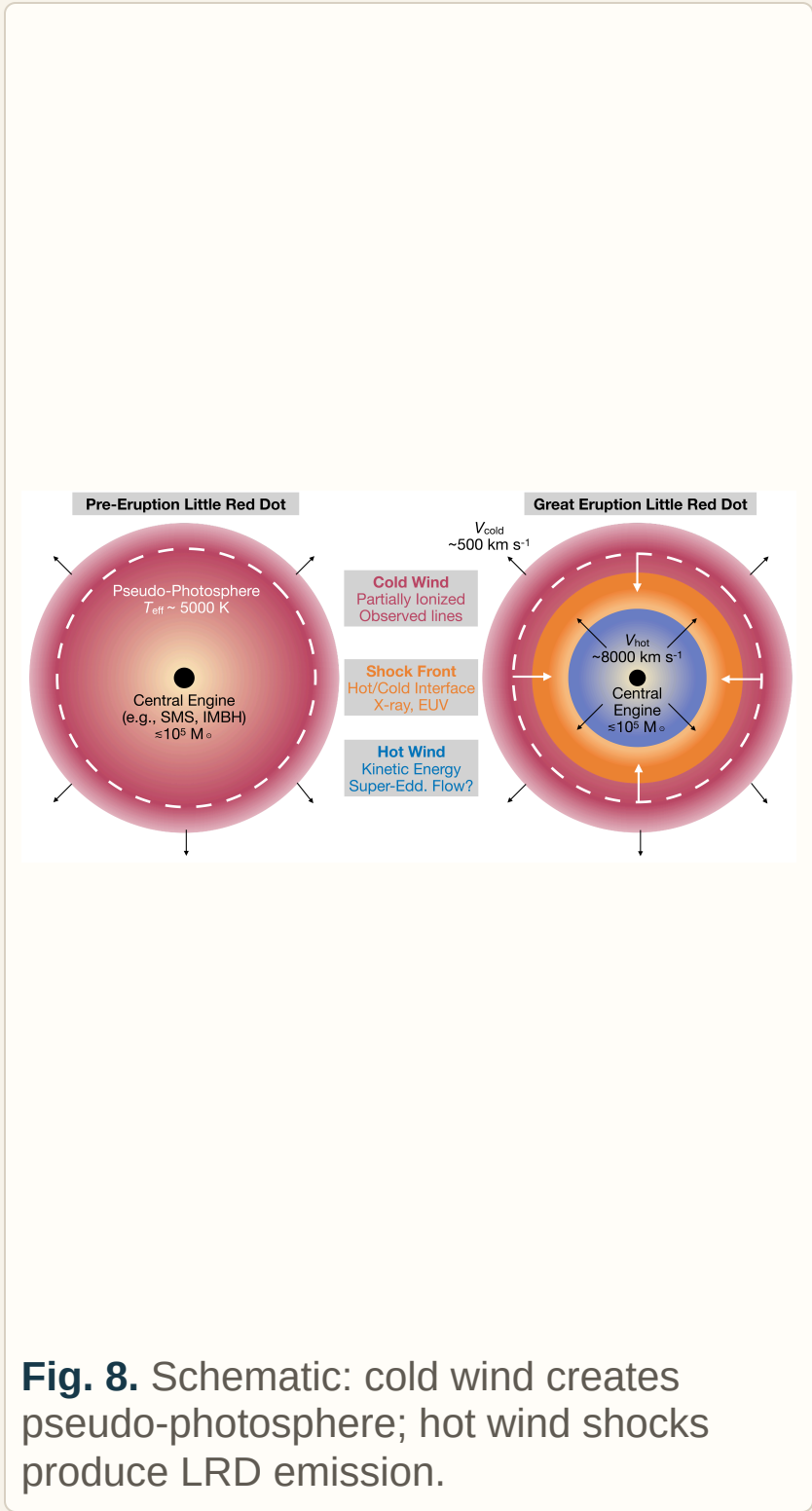


Fig. 8. Schematic: cold wind creates pseudo-photosphere; hot wind shocks produce LRD emission.